

Fats



toppr

Intro of F_{AT}



- ❑ In humans and many animals, fats serve both as energy sources and as stores for energy in excess
- ❑ Each gram of fat when burned or metabolized releases about 9 calories ($37 \text{ kJ} = 8.8 \text{ kcal}$)
- ❑ Fats are also sources of essential fatty acids, an important dietary requirement

Interesting Facts

➤ 1 gram of carbohydrate and Protein gave 04 calorie of energy

➤ But, 1 gram of Fat gave 09 calorie of energy (Double)

Types of Dietary Fats

Unsaturated Fats



Monounsaturated Fats
Canola oil and olive oil.



Polyunsaturated Fats
sunflower oil, and corn oil.

Saturated Fats



Animal meat
Red meat, Lamb,
Pork and Poultry.



Plant Oils
Coconut oil and
Palm kernel.



Processed meat
Hot Dogs
and Sausages.



Dairy products
Milk, Butter
and Cheese.



Snack items
Chips, Cookies
and Pastries.

Trans Fats



Fried foods
French fries, Mozzarella
sticks, and Fish sticks.



**Margarine added
products**
Lots of Bakery items.



**Non-Dairy
Coffee Creamer**
Tea whiteners or
Coffee whiteners.



Meat and Dairy
Especially pork
and Milk.

RDA

Recommended **D**ietary **A**llowance

The UK government recommends that:

- ✓ Men should have less than 30g of saturated fat per day.
- ✓ Women should have less than 20g of saturated fat per day.

Sources

- ❖ “Good” unsaturated fats: Foods high in good fats include vegetable oils (such as olive, canola, sunflower, soy, and corn), nuts, seeds, and fish.
- ❖ “Bad” fats — trans fats: Foods containing trans fats are primarily in processed foods made with trans fat from partially hydrogenated oil.
- ❖ Saturated fats: Foods containing large amounts of saturated fat include red meat, butter, cheese, and ice cream. Some plant-based fats like coconut oil and palm oil are also rich in saturated fat.

Functions

- Carrier of vitamins

In the diet, fat is a carrier for the fat-soluble vitamins A, D, E and K, and supports their absorption in the intestine.

- as a larger, longer-lasting energy reserve

- The body uses fats to produce and control hormones

- The body is protected from severe temperatures by the thick layer of subcutaneous fat

Deficiency

1. Frequent sickness
2. Dermatitis (skin inflammation)
3. Slow wound healing
4. Hair loss

Excess

People with obesity are more likely to develop a number of potentially serious health problems, including:

- Heart disease and strokes
- Type 2 diabetes
- Certain cancers
- Digestive problems
- Sleep apnea
- Osteoarthritis
- Fatty liver disease